ATHARVA SONAWANE

AI/ML Engineer | Full-Stack Developer | Published Researcher

atharvasonawane96@gmail.com | Satara, Maharashtra, India | Portfolio | LinkedIn | GitHub

Summary

AI/ML Engineer specializing in Agentic AI, Generative AI, and Large Language Models, with **three peer-reviewed publications** spanning computer vision, healthcare AI, and retrieval-augmented generation. Experienced building production-grade systems across deep learning, NLP, and computer vision — including an agentic pipeline that cut manual workflow time by **40%** and a face recognition system running at **98% accuracy** with 60% lower inference latency. Seeking full-time AI/ML Engineer, Machine Learning Engineer, or Software Engineer roles.

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, C++, SQL
AI/ML & DL: TensorFlow, PyTorch, Keras, Scikit-learn, Hugging Face Transformers, LangChain, OpenAI API, GROQ
CV & NLP: OpenCV, DeepFace, YOLO, BERT, GPT, Whisper, NER, Sentiment Analysis, Transfer Learning
Web & Backend: React.js, Node.js, Express.js, FastAPI, Flask, REST APIs
Data & Cloud: Pandas, NumPy, MongoDB, PostgreSQL, MySQL, Docker, Kubernetes, AWS (EC2, S3, Lambda, SageMaker), CI/CD, MLOps

Professional Experience

Frontend Developer Intern — Digital Tokri (IRA Digital Services) *Jan 2026*
Wakad, Pune, Maharashtra, India

- Developed and maintained responsive UI components for live web projects using **HTML5, CSS3, and JavaScript** in an on-site Agile environment.
- Collaborated with cross-functional teams on sprint delivery, code reviews, and performance optimization, achieving **95+ Lighthouse scores**.

AWS Generative AI Intern (Virtual) — AICTE – EduSkills / AWS Academy *Jan 2026*
Satara, Maharashtra, India

- Completed a 10-week program covering ML foundations, NLP, sentiment analysis, and **Generative AI with Amazon SageMaker**.
- Built competency in **prompt engineering**, responsible AI practices, and cloud-based ML model development on AWS.

AI/ML/DL & Generative AI Engineer Intern — IOFT *Aug 2025*
Satara, Maharashtra, India

- Architected and deployed production-ready agentic AI systems using reinforcement learning and autonomous agent frameworks, reducing manual intervention by **40%**.
- Engineered generative AI models using transformer architectures and attention mechanisms, achieving a **25% performance improvement**; implemented scalable end-to-end ML pipelines with automated CI/CD.
- Completed the internship **with distinction**, recognized for exceptional technical contributions.

UX Designer & Frontend Developer Intern — Cognifyz Technologies *Nov – Dec 2024*
Satara, Maharashtra, India

- Conducted UX research using analytics, heatmaps, and A/B testing; delivered recommendations that increased user engagement by **30%** and reduced bounce rates.

Key Projects

AI-Powered Healthcare Diagnostic Assistant — Published Research *2024 – 2025*

- Built a multilingual (10+ languages) diagnostic platform combining LLMs, speech recognition (Whisper), and computer vision for real-time medical assessments, using the **GROQ API** for rapid inference and explainable AI for recommendations.
- Published in **IJSAT Journal (2025)**.

Automatic Number Plate Recognition (ANPR) System — Published Research *2024 – 2025*

- Developed a real-time vehicle plate recognition system with CNNs and OCR (Tesseract), reaching **94% detection accuracy, 91% character recognition accuracy**, and 25 FPS video processing.
- Published in **IJRAR Journal (Vol 12, Issue 1, 2025)**.

Intelligent Medical Assistant Framework (LLM + RAG) — Published Research *2025 – 2026*

- Designed a **retrieval-augmented generation (RAG)** framework paired with a large language model to ground medical assistant responses in verified clinical knowledge, reducing hallucinated outputs and improving answer relevance.
- Published as **"Intelligent Medical Assistant Framework Powered by Large Language Model and Retrieval Augmented Generation."**

Multi-Modal AI/ML Desktop Application *2024 – 2025*

- Integrated 6+ AI/ML models (**DeepFace biometric auth at 98% accuracy**, admission prediction at 92%, sentiment analysis) into one GUI with **30 FPS real-time CV** and 60% latency reduction via GPU acceleration.

CipherVault: Enterprise Secure Encryption & Digital Vault *2025 – 2026*

- Built a desktop application for **secure encryption**, confidential note management, encrypted QR code generation, authentication, and activity tracking, using industry-standard cryptography and SQLite.

Smart Lab System (Campus Lab Management) *2025 – 2026*

- Built a full-stack lab automation platform (Django/ASGI + Tkinter kiosk) with real-time floor maps, an **interval-overlap algorithm** preventing double-booking, and WebSocket-based live status sync.

Django REST API Beginner Project *2025*

- Built a **Django REST Framework** API for a blog application with full CRUD operations, custom permission classes, search, and date-based filtering (django-filter).
- Implemented Django's built-in authentication system and used the **DRF browsable API** for testing without external tools.

Education

Bachelor of Technology, Computer Science & Engineering *2022 – 2026*

Yashoda Technical Campus, Wadhe, Satara, Maharashtra • CGPA: 7.0/10.0

Higher Secondary Certificate (HSC) – Science Stream *Jul 2021 – Mar 2022*

Yashwantrao Chavan Institute of Science, Satara, Maharashtra • 75%

Secondary School Certificate (SSC) *Jul 2019 – Mar 2020*

Nirmala Convent High School, Satara, Maharashtra • 86%

Publications

- "AI-Driven Diagnostics: A Multilingual Healthcare Assistant Using LLMs and Computer Vision," IJSAT Journal, 2025.
- "Automatic Number Plate Recognition Using Deep Learning and OCR," IJRAR Journal, Vol. 12, Issue 1, 2025.
- "Intelligent Medical Assistant Framework Powered by Large Language Model and Retrieval Augmented Generation," 2025–2026.

Certifications

- Machine Learning Specialization, Python & Data Science, SQL for Data Analysis, Prompt Engineering for LLMs, AWS Cloud Practitioner Essentials — DataCamp
- Competed in 6+ hackathons/competitions including Kurukshetra 2025 HackFest, HackCelestial 2025, and MHTechin Idea Pitch Deck 2025

Leadership

- President, Cisco Committee — led technical expos, hackathons, and coding competitions for 200+ participants at Yashoda Technical Campus
- Conduct 2 state level hackathons/cometitions leading as the President of the committee.